Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Trachtenberg, Michael Carl Ph.D.		
Project Title:	Multi-purpose Carbon Dioxide Removal System		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08852	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/13/2003	**End Date:**	06/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Goldman, Stefanie (Carbozyme, Inc.) Lihong, Bao (Carbozyme, Inc.)		
Grant/Contract No.:	NAG9-1383		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: slight change in title. We will design, construct, and test an innovative, robust, low mass, low power separation device that can extract carbon dioxide (CO2) from respiratory gases in support of air revitalization for astronaut life support. The research will demonstrate the viability of the separation process by building a proof-of-concept CO2 separator. The separation device will be scaled to 1/8 of an astronaut. The research will focus on optimizing reactor performance, increasing catalyst lifetime, improving reactor design, modeling reactive transport and initiating process engineering designs towards constructing and delivering a demonstration reactor to JSC for later testing. The principal goal of this work was to elevate the status of our (HF-EBCLM) reactor from laboratory development stage to field test, i.e., to increase from TRL 1-3 to TRL 4-6. This will be accomplished by better understanding, controlling		

Task Description:	and optimizing our design. The goals are to: 1. Demonstrate performance commensurate with EVA requirements. i. CO2 permeance of 1.35 x 10-7 moles/m2 s Pa for a feed supply of 1% CO2 ii. CO2 : O2 selectivity of > 2,800:1 2. Demonstrate lifetime of the reactor core for at least 6 months. The Objectives to be satisfied to achieve these goals are: 1. Coordinate three technical areas: (a) CLM chemistry, (b) reactor design, and (c) system engineering. This will be accomplished by: i. Finalize trade-off analysis for the composition of the contained liquid membrane and the enzyme concentration. ii. Complete the 1-dimensional reactive transport (1-DRT) modeling to predict reactor performance. iii. Continue the process engineering effort. 2. Work with NASA (JSC, KSC, ARC) to carry out ESM analysis. NASA JSC will provide in-kind support. This work addresses several Critical Path Roadmap questions including those dealing with control of breathing atmosphere, thermal control and humidity control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The concept is applicable to capture of CO2 as a greenhouse gas to help control possible global warming events.
Task Progress:	1. We developed a model system to predict the preferred chemical makeup of the contained liquid membrane needed to maximize CO2 capture from very dilute feeds such as respiratory gas. 2. We developed exhaustive mathematical models for both chemical reaction, known as equilibrium modeling, and mass transfer, known as rate-based modeling. The models were highly predictive and were validated by experimental measurement. 3. We developed methods for enzyme immobilization in functionalized silica gel and by cross linked enzyme crystals. While these were more stable to denaturants than are solutions of native enzyme the attendant decrease in enzyme activity was significant enough to prevent their use. 4. We developed methods to avoid use of UV sterilization by increasing salt content. While modest, this level is sufficient to prevent bacterial growth in the reactor. 5. We demonstrated stability of the reactor for a period of at least 5-days of continuous run. 6 . Overall system performance improved from 10% of that needed for EVA to about 50% and met certain goals for ALS applications. 7. We carried out process engineering to initiate design of a piece of apparatus for EVA applications. 8. We designed a next generation to improve performance as much as 20-fold.
Bibliography Type:	Description: (Last Updated: 12/01/2010)
Abstracts for Journals and Proceedings	Trachtenberg MC, Bao L, Goldman SL, Smith DA, Wu X. "Dynamic maintenance of CO2 levels in closed environments." NASA JSC meeting, November 13, 2004. NASA JSC meeting, November 2004. , Nov-2004
Abstracts for Journals and Proceedings	Trachtenberg MC, Bao L, Goldman SL. "CO2 transfer across liquid membrane facilitated by carbonic anhydrase. " AIChE Annual Meeting, Austin, TX, November 7-12, 2004. AIChE Annual Meeting, Austin, TX, 2004. Abstract #241F. , Nov-2004
Abstracts for Journals and Proceedings	Trachtenberg MC, Bao L, Goldman SL. "Flue gas CO2 capture by a green liquid membrane. " GHGT-7, Vancouver, BC, September 7-11, 2004. GHGT-7 meeting, September 2004. , Sep-2004
Abstracts for Journals and Proceedings	Trachtenberg MC, Bao L, Smith DA, Goldman SL. "Air revitalization in ALS and EVA." Advanced Life Support Air Revitalization Element Workshop, Gainesville, FL, Jan. 8-9, 2004. Advanced Life Support Air Revitalization Element Workshop, January 2004. , Jan-2004
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